

KRISHNA CHARAN THOTA

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SUMMARY

Software Engineer with 4+ years building scalable backend systems and cloud-native services using Go, Java, AWS, and Kubernetes. Worked on distributed ingestion pipelines, secure webhook delivery platforms, observability systems, and large-scale data processing across Amazon and Armada. Strong experience in debugging production issues, improving system reliability, and owning services end-to-end in fast-paced environments.

PROFESSIONAL EXPERIENCE

Software Engineer, Armada AI, Bellevue, WA

Mar 2025 – Present

- Designed and delivered an end-to-end cloud-native data ingestion pipeline for private 5G and satellite infrastructure, integrating **Nokia** and **Starlink** vendor APIs into a unified analytics platform covering radio access nodes, SIM cards, fault alarms, and performance metrics at scale.
- Built a distributed webhook delivery system integrating with third-party platforms like **ServiceNow** via **JWT**-secured authentication and **HMAC** payload signing, enabling structured event delivery as an alternative to high-volume email alerts.
- Resolved a critical silent data loss issue where edge device alarms were dropped during ingestion due to missing asset associations, implementing a **fault-tolerant fallback** that eliminated all skipped alarms in production.
- Optimized ingestion throughput using **async job orchestration** and **worker pool processing**, improving reliability across distributed edge and IoT device ingestion at scale.
- Built SIM lifecycle management APIs across multiple service layers, handling vendor-specific edge cases including partial success states and inconsistent **SIM synchronization** issues in production.
- Implemented **exponential backoff retry**, async goroutine dispatch, and **ScyllaDB**-backed delivery tracking for webhook reliability, with automated Mailgun alerts on retry exhaustion.
- Built **Prometheus** and **Grafana** observability dashboards covering P99 latency, vendor success rates, in-flight job tracking, and pod-level resource utilization across distributed services.
- Integrated **Azure Key Vault** with Managed Identity for secret management and secure token retrieval workflows, eliminating hardcoded credentials and strengthening security compliance across services.
- Implemented automated deployment rollback and health check validation across **CI/CD** pipelines, reducing failed deployments and improving release safety.
- Implemented audit logging across the platform covering **webhook** lifecycle events and **SIM provision, deprovision, and edit** actions, providing full traceability for compliance and governance requirements.
- Authored full **OpenAPI/Swagger** documentation covering dual versioned API base paths across multiple services, reducing integration friction for cross-functional teams.
- Owned production incident response as primary **on-call** engineer, driving high-severity issue resolution and cross-team escalation coordination to maintain service reliability.

Software Development Engineer, Amazon Services LLC, Bellevue, WA

Oct 2022 – Mar 2025

- Designed a high-performance **Java** API integrated with **DynamoDB**, reducing data retrieval times by **29%** and improving operational efficiency through structured design patterns and best practices.
- Architected **ETL** processes for **1.2 billion** metadata records, consolidating data into a centralized **Redshift** data warehouse, reducing reporting time by **27%** and improving decision-making accuracy.
- Designed a **pre-production** environment with AWS **CodePipeline** and **CDK** in **TypeScript** to validate infrastructure changes before production, minimizing risks through automated monitoring and logging.
- Executed a **throttling** configuration **limiting concurrent** requests during peak traffic, resulting in a **95%** reduction in service disruptions and preventing cascading failures under high load.
- Leveraged AWS services including **API Gateway**, **S3**, **SQS**, **CloudWatch**, **CloudFormation**, **IAM**, **ECS**, and **Fargate** to design, develop, and deploy scalable cloud-native applications.
- Led full **SDLC** across requirement **gathering**, **design**, **testing**, and **deployment** for high-priority **Java** applications, achieving a **98%** on-time delivery rate and consistently meeting delivery timelines across cross-functional teams.
- Drove continuous process improvement as **Scrum Master**, improving team productivity by **15%** through Agile practices and actively contributing to **on-call** rotations.
- Engineered modular Java microservices using **OOP** principles and **Spring Boot**, resulting in a **30%** reduction in debugging time across critical software projects with seamless **NoSQL** database integrations.
- Promoted code quality and maintainability through structured **unit testing** with **JUnit** and **Mockito**, achieving high test coverage across critical service components.

Software Dev Intern, AbhiBus Services India Private Limited, India

Dec 2019 – Apr 2020

- Implemented a robust **Python** application featuring a **HTML** and **CSS** front-end to display extracted **competitor data**, resulting in enhanced comparative analysis capabilities and reducing manual research time by 50 hours each month.
- Engineered a high-performing **machine learning** algorithm in **Python** with a 90.4% accuracy rate for predictive analysis, seamlessly integrating it into the **React.js** for real-time data prediction and display.

- Conducted **factor analysis** and integrated the results into the **React.js** user interface, providing **intuitive data visualization** components for effective **interpretation** and strategic insights.

SKILLS

Programming Languages: Java, Go, Python, TypeScript, JavaScript, Bash, HTML, CSS
Cloud & Databases: AWS (API Gateway, S3, SQS, CloudWatch, CloudFormation, IAM, ECS, Fargate, EC2, CDK), DynamoDB, Redshift, PostgreSQL, MySQL, MongoDB, Oracle, ScyllaDB, Azure (Key Vault, Managed Identity, Workload Identity, AKS)
Frameworks & Libraries: Spring, Spring Boot, Spring Cloud, JUnit, Mockito, React.js, Redux, Jest, Cypress
Observability & Monitoring: Prometheus, Grafana, CloudWatch, Structured Logging, OpenAPI/Swagger
Security & Authentication: JWT (RS256), OAuth2, HMAC, Keycloak, Workload Identity Federation, Azure Key Vault
Containerization: Docker, Kubernetes, Helm, Istio
Development Tools: Git, GitHub, Maven, Jenkins, Jira, Asana, Agile/Scrum, CI/CD, QuickSight
APIs & Architecture: REST, gRPC, Microservices, Distributed Systems
Operating Systems: Linux, Windows

CERTIFICATES

- [AWS Solutions Architect Associate](#)

EDUCATION

Masters of Business Administration, WestCliff University	Sep 2025 – Present
Master of Professional Studies in Data Sciences and Applications, University at Buffalo, State University of New York	Aug 2021 – Aug 2022
Bachelors, Computer Science and Engineering, Vignan’s Foundation for Science Technology and Research	May 2016 – May 2020